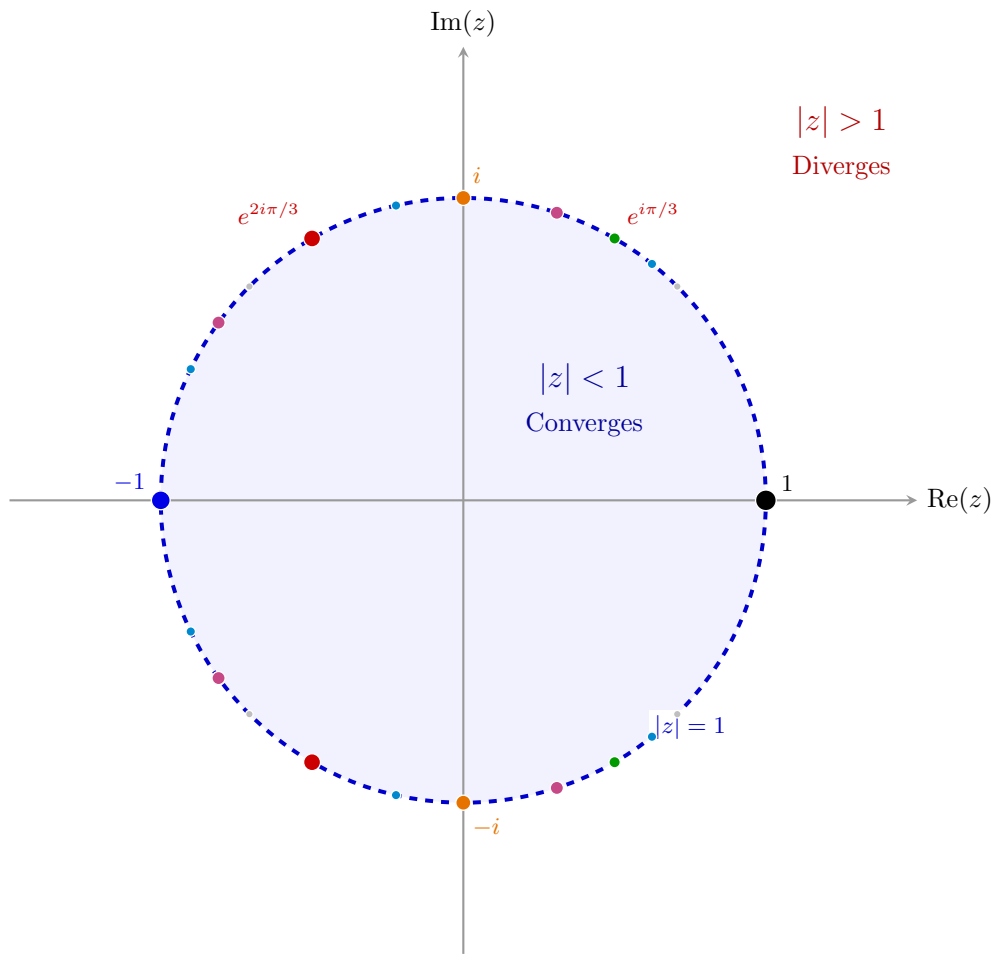




Complex Plane: Roots of Unity & The Natural Boundary

The zeros of $\phi(z) = \prod_{n=1}^{\infty} (1 - z^n)$ occur exactly at every n -th root of unity.

As $n \rightarrow \infty$, these zeros form a completely dense "wall" of singularities along the unit circle, creating a **natural boundary** beyond which the function cannot be analytically continued.